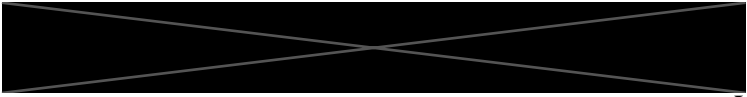
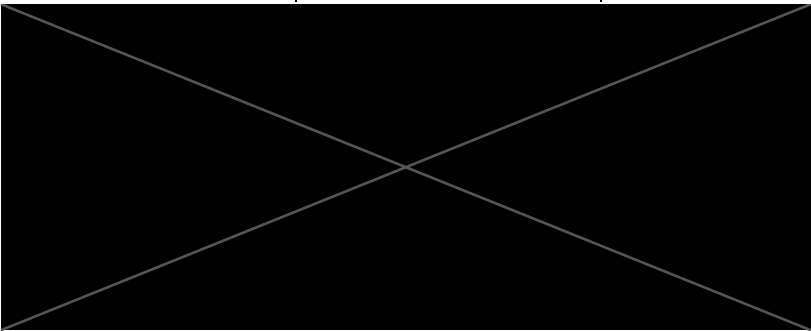


|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>UMS</b><br>UNIVERSITI MALAYSIA SABAH<br> <b>FACULTY OF COMPUTING &amp; INFORMATICS</b><br>FAKULTI KOMPUTERAN DAN INFORMATIK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>BORANG DEKLARASI (BERKUMPULAN) / DECLARATION FORM (GROUP)</b><br><b>UNTUK PENILAIAN MOD ASYNCHRONOUS / FOR ASYNCHRONOUS</b><br><b>MODE ASSESSMENT</b>                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>A. MAKLUMAT PELAJAR/STUDENT'S INFORMATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Nama Penuh Ketua / Leader's Full Name:<br>[Mengikut MyKad atau paspot / As in MyKad or Passport]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Syazra Syazana binti Anwar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| No. Matrik Ketua / Leader's Matric No.:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| No. Telefon dan e-mel ketua / Leader's Phone No. and e-mail:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Nama Kursus / Course Name:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | System Analysis and Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Kod Kursus / Course Code:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | KP24102                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Jenis Penilaian / Assessment Type :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;"> <input checked="" type="checkbox"/> Tugas / Assignment         </div> <div style="width: 50%;"> <input type="checkbox"/> Latihan Makmal / Lab Exercise         </div> <div style="width: 50%;"> <input type="checkbox"/> Kuiz / Quiz         </div> <div style="width: 50%;"> <input type="checkbox"/> Projek / Project         </div> <div style="width: 50%;"> <input type="checkbox"/> Lain-lain / Others : (Nyatakan / Specify)         </div> </div> <p>.....</p> |
| <b>B. DEKLARASI/ DECLARATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>SILA BACA DENGAN TELITI SEBELUM MELENGKAPKAN BORANG INI/ PLEASE READ CAREFULLY BEFORE COMPLETING THE FORM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <ol style="list-style-type: none"> <li>Saya dengan ini mengaku bahawa penilaian ini adalah hasil kerja saya sendiri sepenuhnya. Sebarang sumber maklumat tambahan, sama ada diterbitkan secara terbuka atau tidak diterbitkan, yang telah digunakan dan dirujuk dalam teks, senarai kandungan dan/atau senarai bibliografi telah diakui dengan sewajarnya dan semakan kesamaan melalui Turnitin adalah tidak melebihi 30%.<br/> <i>I hereby declare that this assessment is entirely my own work. Any additional sources of information, whether publicly published or unpublished, that have been utilized and referenced within the text, contents list, and/or bibliographical list have been duly acknowledged and similarity check through Turnitin is not more than 30%</i> </li> <li>Kami tidak akan membenarkan sesiapa menyalin atau mengeluarkan semula karya kami tanpa kebenaran kami dengan mengambil semua langkah berjaga-jaga.<br/> <i>We will not allow anyone to copy or reproduce our work without our authorization by taking necessary precautions.</i> </li> <li>Kami sedar dan memahami dasar Universiti mengenai plagiarisme dan amalan akademik yang baik. Kami sedar sepenuhnya bahawa sebarang pelanggaran dasar yang berkaitan akan mengakibatkan peruntukan sifar markah untuk penilaian ini dan mungkin akan mengakibatkan tindakan tatatertib.<br/> <i>We are aware of and understand the University's policy on plagiarism and good academic practices. We are fully aware that any violation of the relevant policies will result in the allocation of zero marks for this assessment and may also warrant disciplinary actions.</i> </li> </ol> <div style="margin-top: 10px;"> <input checked="" type="checkbox"/> Saya ada menggunakan AI tools dalam penyelesaian tugas/ penilaian ini<br/>       I have used AI tools in this assignment Assessment     </div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>C. PERAKUAN PELAJAR / STUDENT DECLARATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| No. K.P. / Paspot Ketua: <br>Leader's I.C. no. / Passport :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Tandatangan Ketua / Leaders's Signature:<br><div style="display: flex; align-items: center; margin-top: 20px;">  <div style="margin-left: 20px;">           Tarikh / Date: 01 01 2024         </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

**Nota/ Notes**

- Pelajar adalah digalakkan untuk melampirkan bersama laporan Turnitin (Kurang 30%) bersama borang ini tertakluk kepada permintaan pensyarah masing-masing.  
 Students are encouraged to attach their Turnitin reports (Less than 30%) with this form subject to the lecturer's request.

|  <b>UMS</b><br>UNIVERSITI MALAYSIA SABAH |                                   |  <b>FACULTY OF COMPUTING &amp; INFORMATICS</b><br>FAKULTI KOMPUTERAN DAN INFORMATIK |                                     | <b>BORANG DEKLARASI (BERKUMPULAN) / DECLARATION FORM (GROUP)</b><br><b>ASSIGNMENT/ LAB EXERCISE/ QUIZ/ PROJECT</b> |  |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------|--|
| <b>D. SENARAI AHLI KUMPULAN / LIST OF GROUP MEMBERS</b>                                                                  |                                   |                                                                                                                                                                      |                                     |                                                                                                                    |  |
| NO.                                                                                                                      | NAMA / NAME                       | NO. MATRIKS /<br>MATRICS NUMBER                                                                                                                                      | NO. KAD PENGENALAN /<br>I.C. NUMBER | TANDATANGAN / SIGNATURE                                                                                            |  |
| 1.                                                                                                                       | Syaza Syazana binti Anwar         |                                                                                    |                                     |                                                                                                                    |  |
| 2.                                                                                                                       | Norsyiah binti Safudin            |                                                                                                                                                                      |                                     |                                                                                                                    |  |
| 3.                                                                                                                       | Wan Nuradliana Balais Bt Wan Azmi |                                                                                                                                                                      |                                     |                                                                                                                    |  |
| 4.                                                                                                                       | Shwentta Parani Kumar             |                                                                                                                                                                      |                                     |                                                                                                                    |  |
| 5.                                                                                                                       | Leena A/P Seliamany               |                                                                                                                                                                      |                                     |                                                                                                                    |  |
| 6.                                                                                                                       |                                   |                                                                                                                                                                      |                                     |                                                                                                                    |  |
| 7.                                                                                                                       |                                   |                                                                                                                                                                      |                                     |                                                                                                                    |  |
| 8.                                                                                                                       |                                   |                                                                                                                                                                      |                                     |                                                                                                                    |  |
| 9.                                                                                                                       |                                   |                                                                                                                                                                      |                                     |                                                                                                                    |  |
| 10.                                                                                                                      |                                   |                                                                                                                                                                      |                                     |                                                                                                                    |  |



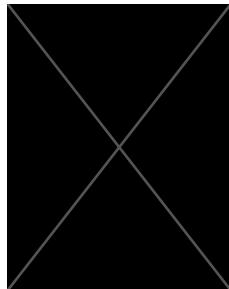
**UMS**  
UNIVERSITI MALAYSIA SABAH

**KP24102 SYSTEM ANALYSIS AND DESIGN**  
**SEMESTER 1, 2023/2024**

**LECTURER: DR LAI PO HUNG**

**PROJECT REPORT:**  
**MARATHON TRAINING APP**

**PREPARED BY:**

| <b>BIL</b> | <b>NAME</b>                       | <b>MATRIC NUM</b>                                                                     |
|------------|-----------------------------------|---------------------------------------------------------------------------------------|
| <b>1</b>   | SYAZA SYAZANA BINTI ANWAR         |  |
| <b>2</b>   | SHWEHTTA PARANI KUMAR             |                                                                                       |
| <b>3</b>   | NORSYLAH BINTI SAFRUDIN           |                                                                                       |
| <b>4</b>   | WAN NURADRIANA BALQIS BT WAN AZMI |                                                                                       |
| <b>5</b>   | LEENA A/P SELLAMANY               |                                                                                       |

**TASK ALLOCATION:**

1. Syaza Syazana binti Anwar: Report Writing, Class Diagram, Interfaces Design
2. Shwetta Parani Kumar: Report Writing, Context Diagram, Interfaces Design
3. Norsyiah binti Safrudin: Report Writing, Data Flow Diagram, Interfaces Design
4. Wan Nuradriana Balqis bt Wan Azmi: Report Writing, Class Diagram, Interfaces Design
5. Leena A/P Sellamany: Report Writing, Entity Relationship Diagram, Interfaces Design

## **TABLE OF CONTENTS**

|                                            |    |
|--------------------------------------------|----|
| 1.0 Purpose of This System                 | 3  |
| 2.0 What Problems Will it Solve            | 5  |
| 3.1 Marathon Announcement and Registration | 6  |
| 3.2 Data Synchronization                   | 6  |
| 3.3 Structured Marathon Guidance           | 7  |
| 3.4 Progress Tracking and Analysis         | 8  |
| 3.5 Community and Virtual Coach Support    | 8  |
| 4.0 Methods to Gather Information          | 10 |
| 5.0 System Requirements                    | 11 |
| 5.1 Hardware Requirements                  | 11 |
| 5.2 Software Requirements                  | 11 |
| 5.3 Communication Protocols                | 12 |
| 6.0 Tools of Information Gathering         | 14 |
| 6.1 Google Forms                           | 14 |
| 6.2 Comparison Items                       | 18 |
| 7.0 References                             | 20 |

## **1.0 Purpose of This System**

The Marathon Training Apps now become a huge hit in the fitness industry. This technological app aims to improve and streamline the marathon training process. These apps are designed to empower all runners making the journey to the marathon finish line more enjoyable and accessible. In order to understand why this training app has now become the game changer to the runners alike we will examine the purpose that makes these apps essential components in a runners journey towards marathon mastery. Initially, the runners should have their training before the marathon as the marathon demands physical activity that calls for muscular strength, cardiovascular health and endurance. Appropriate training will help runners prepare their bodies for the demands of running a long distance. Before designing the system, we point out the needs of runners so that we can provide the purpose of it.

The runners need a centralized source of information about upcoming marathons to plan their race calendar. Besides, the purpose of this system is to provide a comprehensive list of marathon events, including the dates and locations. It also makes secure registration easier for users, making the process more efficient.

Moreover, the runners also need professional training programmes that take into account their schedules, goals and fitness level. That is the purpose of this system which fulfills this need by offering structured and customizable training plans that provide runners with clear guidance and systematic approach to their marathon preparation.

On the other hand, runners need to track their progress and make wise decisions, so what they need is a real time tracking and analysis of their performance. As a result, the purpose of the system is to fill this gap by providing tools for monitoring distance, pace and other relevant information. It also offers analytical tools that provide information on performance patterns and potential areas for development.

Furthermore, we believe that in order to gain a full understanding of their health, runners want a common understanding of their fitness and health data from multiple sources. The purpose of this system is to make data synchronization and integration. This app should seamlessly sync with the popular fitness tracker to integrate data and provide users a comprehensive understanding of their overall health and fitness. We also wish to provide real time data updates to ensure accuracy and relevance in the training journey.

Lastly, during solo training sessions participants might feel unmotivated and unsupported, the lack of it makes the purpose of this system by providing community and

virtual coach support, this is enabling users as the runners to interact with other runners through social features. The addition of the virtual coach also helps runners to be more dedicated as the virtual couch provides expert advice and encouragement, addressing the need for guidance and assistance.

## **2.0 What Problems Will it Solve**

### **Repeated Exercise Programs:**

**Problem:** Boredom and decreased engagement may result from repetitive workouts that make marathon training monotonous.

**Solution:** The system will analyze the workout pattern of users to produce a variety of workouts to try. This ensures the users enjoy the journey of the marathon.

### **Insufficient Customization in Training Programs:**

**Problem:** Individual fitness levels, objectives, and preferences may not be catered for in generic training plans, which could result in less than ideal outcomes.

**Solution:** The app's personalized virtual coaching features provide individualized advice based on users' performance, goals, and progress. By doing this, training programs are made more effective by ensuring that their plans are flexible and tailored to each participant's needs.

### **Restricted insights based on data:**

**Problem:** Users might not have access to useful information about how their training is going, which would prevent them from making wise decisions.

**Solution:** The combination of wearable technology and real-time data analytics offers users a comprehensive understanding of their performance, enabling them to monitor their progress, pinpoint areas in need of development, and make informed decisions to enhance their training regimens.

With the help of community and virtual support features, marathon training apps seek to solve these issues and increase user engagement, training results, and the general well-being of those who are aiming for marathons.

### **3.0 Scope**

#### **3.1 Marathon Announcement and Registration**

This marathon announcement and registration is one of the main scope in our Marathon Training app. This scope focuses on dynamic announcements for the user and an easy registration procedure. The main features of this is its dynamic real-time marathon announcement which sends out personalized notifications about upcoming events based on user preferences, by creating a sense of motivation and anticipation. This scope offers the user more than just an announcement, it also lets them personalize their experience running a marathon by sending relevant information about the specific challenges related to the marathon. In addition, this app provides a simple and quick registration process. By registering for their preferred marathons with ease, runners will be able to get away with the difficulty that comes with registering for events. Moreover, for the long journey, the app adds interesting user profiles. By entering their fitness level and information about themselves this personalized reach guarantees that users or runners are ready for the unique demands of the selected events and each time the events that comply with their criteria the user will get the announcement or the notification. To sum up, by offering this real-time notifications, effortless registration process and engaging user profile, this app not only meets the needs of the running community right away but also helps ensure the long term stability and satisfaction of runners everywhere. This app serves as a motivator and one of the factors contributing to the runners success journey in the marathon.

#### **3.2 Data Synchronization**

Data synchronization is proven to be one of the most important features in every app ever. Data synchronization is a process used in applications to guarantee that data is current and consistent across various systems or devices. Through observations and experience, people find it annoying to just be able to get info in one certain device or system and need to transfer it manually to another device. When having data synchronization, users can have many devices with the same data available in every device included automatically. Thus, we included data synchronization as one of the features provided by our Marathon Training apps. Users of marathon apps usually sync their smartwatch and phone to check their progress. Smart watches also make it easier for runners because they only have to put it on their wrist rather than using



a phone where they need to hold on to it all the time. Smart watches to track movements and phones to summarize all the activities done by users. Without data synchronization, runners would not be able to use smartwatches to track their movements.

Apart from that, data synchronization also allows users to use the apps when having poor connectivity. In the case of runners who like to run in the forest or any place with no internet connectivity, then it would be hard to track their progress. Thus, users can work offline thanks to data synchronization, which saves data locally when progresses are made. The app syncs with the server, updating and reconciling changes, as soon as connection is restored. Not only that, data synchronization also ensures data back up. In the case of accidental deletions or device damages or any causes that lead to data loss, then data synchronization will come to help. This is because our Marathon training app can act as a backup by syncing data to a cloud storage account or a central server. Users can restore their data from the synced copy, then everything will be back per normal. Besides, data synchronization assures cross-platform compatibility. Users might have an iOS operated smartphone and an Android powered smartwatch or other devices. In this case, data synchronization will be used to make sure our Marathon Training apps are able to be made of use in every device and its data aligned with each other. Thus, this will make sure users have a seamless experience using our app regardless of what device they are using.

In summary, data synchronization is a feature that can guarantee a smooth and responsive experience when using our Marathon Training application as what they are expecting. A responsive user interface can be preserved by synchronizing data in the background. The most recent data is displayed to users instantaneously, improving their overall experience. Thus, it is crucial to keep data synchronized and make our app a modernized system with no delays and reliable to our respected users.

### **3.3 Structured Marathon Guidance**

The structured marathon guidance included as one of the scope in the marathon training application presents a comprehensive approach to preparing individuals for the physical demands of a marathon. The training module of the app can provide customized training plans based on the user's goals, running experience, and level of fitness. To maximize endurance and performance, it consists of a variety of runs, including long runs, interval training, and speed workouts, all of which are gradually advanced. The app's all-encompassing strategy, which

involves planning thorough and structured training plans, helps prospective marathoners not only prepare physically but also lead a sustainable and healthy lifestyle.

### **3.4 Progress Tracking and Analysis**

Progress tracking and analysis is one of the features that is included in the development of Marathon Training Application. It enhances user experiences, keeps user engagements and more. Progress tracking and analysis feature enables the users to set running goals such as the distance covered on a daily as well as weekly which leads to comparative analysis to compare their current running performances with the past ones. Moreover, training load and recovery insights built in the app will ensure that the user can balance between their training and rest period in order to avoid overtraining and reducing the risk of injuries. Apart from that, this feature also shows performance metrics where it helps the users to recognise their strengths and the areas to improve the next time. In the end, the feature enables to show the overall visual progress report which is represented as charts and graphs. Therefore, this feature is beneficial for the runners to keep themselves on track with their training prior to their marathon.

### **3.5 Community and Virtual Coach Support**

The emphasis on holistic well-being is growing in the fitness industry as it evolves. In order to improve user engagement within a fitness system, this report investigates the integration of community and virtual support. Fitness systems can offer a more comprehensive and customized experience by leveraging virtual tools and cultivating a sense of community. This can lead to increased user satisfaction and sustained adherence to fitness regimens. Since communities and online support have been incorporated into marathon training apps, the marathon training scene has undergone a tremendous change. Acknowledging their potential to transform the conventional method of marathon training, this report investigates how these features can be integrated. Marathon training apps can encourage users to be more motivated and accountable by creating a sense of community among them. Further improving the overall effectiveness of these apps is the integration of virtual support features like real-time progress tracking, individualized coaching, and immersive training sessions. With a focus on psychological and emotional well-being in addition to the physical aspects of training, this all-encompassing

method creates a supportive environment that promotes sustained engagement and success in marathon sports.

The user experience, technical infrastructure, and scalability of marathon training apps must all be carefully considered for the successful integration of community and virtual support. Case studies provide examples of how these features have improved performance outcomes and user engagement noticeably. The combination of these components, which range from online group runs to individualized coaching sessions given via the app, produces a dynamic and customized training environment. Better training compliance is just one result of the influence; another is the emergence of a thriving community of marathon runners who exchange stories and offer encouragement to one another. These marathon training applications are continuously assessed using key performance indicators and user feedback, which guarantees that they adapt to changing user preferences and maintain their position as essential resources in the modern marathon training environment.

#### **4.0 Methods to Gather Information**

In order to gather the crucial information prior to the development of our Marathon Training Application, we have implemented two methods including observation and a survey using Google form. Both the methods used are to collect data to develop a better Marathon Training Application for the future users. We have observed several current applications and seek for the lacking in the current apps. Hence, as a way to gather information to develop an app with better and more useful features, a survey is conducted.

The survey is conducted in the form of a questionnaire designed to collect the insights from potential users of our Marathon Training Application such as their preferences and challenges from such applications. A google form which contains a mix of questions such as checkboxes, close-ended and scales is created. For example, the checkbox type of questions asked is helpful in a sense that it would lead to rich data collection and flexibility as well. The questionnaire makes sure that the data gathered is both quantitative and qualitative. This ensures the analysis made out of it provides comprehensive analysis statistically and descriptively.

We allow the respondents to provide anonymous responses since we prioritize user-friendliness and accessibility of the respondents. The google form is distributed on online platforms such as social media including Whatsapp and Instagram to engage more participants to gather more holistic information needed to develop the Marathon Training Application. As the data is collected from the survey in the form of Google form ongoing data analysis will be conducted to identify the patterns and trends of the responses of the participants of our survey.

Therefore, the findings from the survey we have conducted will be contributory in creating the key aspects of our project which includes the user needs, features etc that are crucial in developing the Marathon Training Application. Apart from that, the methods used such as the survey in the form of questionnaires helps us to come up with better and futuristic features which could be helpful for the users in the future where the Marathon Training Application will always be on demand among the users. Moreover, these methods used are also useful to propose more updates on the application developed. Finally, the insights gathered will serve as a guide in developing a user-centric Marathon Training Application.

## **5.0 System Requirements**

### **5.1 Hardware Requirements**

- **Server**
  - The server is required to have adequate processing power and memory so that it can accommodate simultaneous user connections and for a smooth app.
- **Devices**
  - Available for standard smartphones (iOS and Android), tablets, and compatible smartwatches with internet connectivity.
  - GPS capability for accurate tracking of running routes.
  - Accelerometer support for step counting and movement tracking.
- **Smartwatches**
  - Compatibility with popular smartwatch platforms such as watchOS for Apple Watch and Wear OS for Android Wear
  - Sensors for tracking biometric data, such as heart rate.

### **5.2 Software Requirements**

- Server-side scripting: PHP, Python, or similar technology.
- Database: MySQL, PostgreSQL, or another relational database.
- Web server: Compatible with Apache, Nginx, or equivalent.
- Mobile App Development: Swift for iOS, Kotlin/Java for Android.

#### **Mobile Devices (iOS and Android):**

- Operating System: iOS 12.0 or later (for iPhone) and Android 7.0 (Nougat) or later.
- App Development: Developed using Swift for iOS and Kotlin/Java for Android.

#### **Smartwatches**

- Compatibility with respective smartwatch operating systems (e.g., watchOS, Wear OS).
- Integration with health and fitness tracking APIs.

## **Network Requirements**

- Reliable internet connection for both server and clients.
- Secure sockets layer (SSL) for data encryption.

## **Development Tools**

- Integrated Development Environment (IDE) for iOS and Android development.
- Version control system for collaborative development.

## **Security Measures**

- Encryption protocols to ensure the security of user data.
- Robust user authentication and authorization mechanisms.
- Regular security audits and updates.

## **Audio-Video Components**

- Integration with device audio for feedback or coaching.
- If incorporating live coaching sessions, ensure compatibility with device cameras, microphones, and speakers.

## **Scalability**

- Design the system to handle potential growth in users, including support for smartwatch users.

## **User Interface (UI) and User Experience (UX)**

- Intuitive and user-friendly interfaces tailored for smartphones and smartwatches.

## **5.3 Communication Protocols**

- Define communication methods, such as in-app messaging and notifications, suitable for both mobile devices and smartwatches.
- Compatibility with optional external devices like Bluetooth headphones.

## **5.4 Backup and Recovery**

- Implement regular backups of user data and establish a robust recovery plan in case of system failures.

## 6.0 Tools of Information Gathering

### 6.1 Google Forms

#### MARATHON TRAINING APPS 🏃

Hi! We are from UH6481002 UMS Network Engineering, conducting a survey regarding your preferences about a marathon training apps 🙌🏻

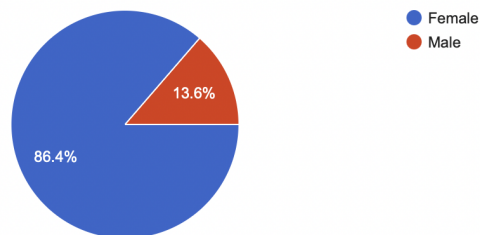
Marathon training apps are mobile applications designed to assist individuals in preparing for and successfully completing a marathon, which is a long-distance running race typically covering 26.2 miles (42.195 kilometers).

Help us enhance your training experience by sharing your thoughts through this quick survey. Your valuable feedback will shape the future of our app, ensuring an effective training companion for every runner 🏃,

Let your voice be heard, and together, let's stride towards excellence in marathon training! 🙌🏻

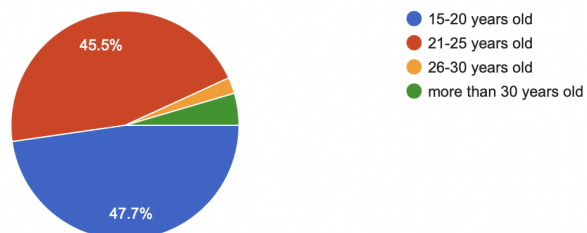
##### Gender

44 responses



##### How old are you?

44 responses

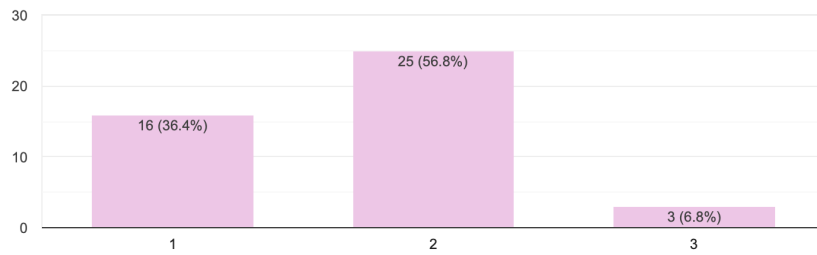




How often do you go for a long-distance running?

[Copy](#)

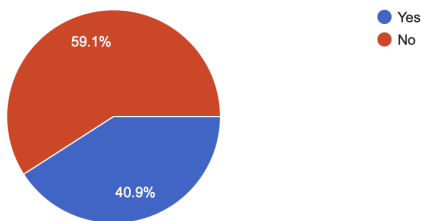
44 responses



Do you ever use a Marathon Training apps? e.g Strava, Runkeeper, Nike Run Club etc

[Copy](#)

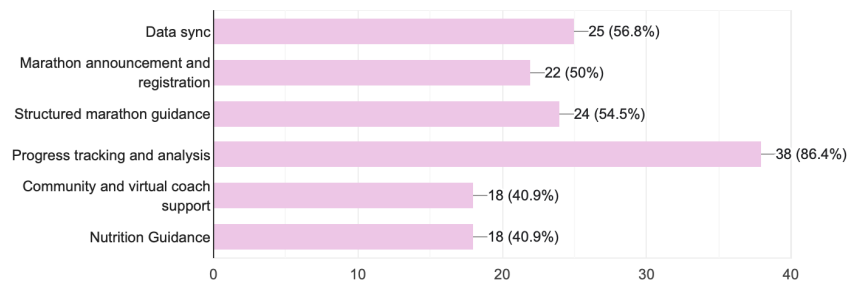
44 responses



Which features below you think will help the most when using a Marathon Training apps? (choose 3-5)

[Copy](#)

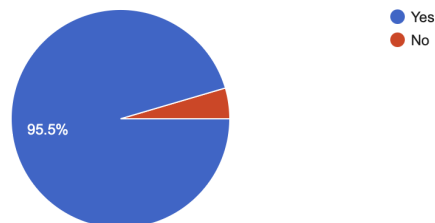
44 responses



Will tracking the progresses of marathons encourage runners to run more?

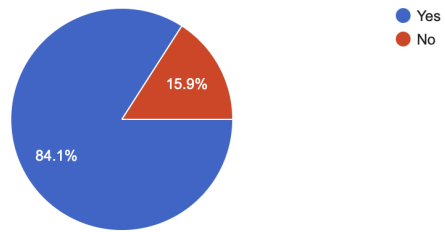
[Copy](#)

44 responses



If there are any marathon events conducted, would you like to participate?

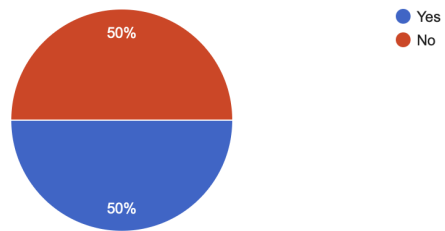
44 responses



---

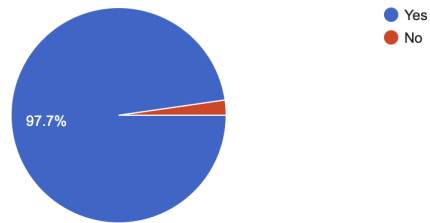
Do you use any smart tools to keep track of your running pace? e.g smart watch etc

44 responses



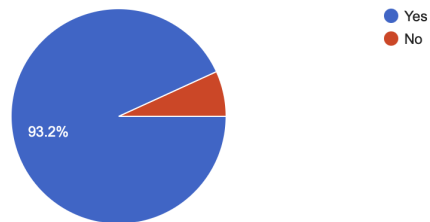
Do you think a community support will help in achieving goals in marathon running?

44 responses



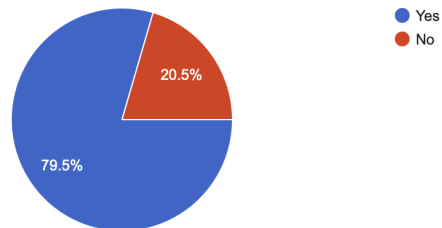
Do you usually warm up your body before working out?

44 responses



Will a virtual marathon coach be any useful to coach and motivate?

44 responses



## 6.2 Comparison Items

We compared several marathon applications and saw what can be improved.

### Huge problem

29 Sep

★★★★☆

nicelyg

You know, i was using this app for personal goals. It is a great app, simple and nice but there was an issue. The map is very buggy, they couldnt accurately locate what line i ran and because of that. Everytime i ran 1km, i can see that the distance changes further overtime and it caused me to be probably 2 minutes slower than the actual recorded time i should have done. Please fix this

### Cannot update running activity

29 Dec

★★★★☆

EncikRiderKental

Since last week, we can't update running activity into strava from Zepp apps. Other than that, still able to update other activity eg workout. Only iphone user has the problem so far. Kindly fix it.

### How about the iPad?

5y ago

★★★★☆

Gilbertenstein

How many years before an iPad version comes out?

Record on the phone. Analyse on the iPad

The iPad version is just a low res view of the phone.

Seems like your missing out on potential features right there. No one uses a PC any [more](#)

**Not stable after recent update** 4y ago  
★★★★☆ Syaeerah

I have been using NRC since 2013 and it is really a great app! Starting from last year, I synced the runs with my Apple Watch. It was doing fine until the recent update of iOS 13 and Watch OS6. The run DID NOT sync AT ALL from my Apple Watch to my phone. I ran only using my watch, and usually whenever I open the app on my phone, the recent run will appear and sync. The app works just fine when I run, I did press STOP and the run ended just fine. But it didn't appear in my recent activity on both my Apple Watch and iPhone. What happened?

**Guided run not played on AirPods & Phone and Watch Not Sync** 11 Oct  
★★★★☆ Hsiendiehdj

Whenever I'm using my Apple Watch to play any Guided Run, the voice assistant will not play through my AirPods . It's really annoying because my music can be played through AirPods but not the NRC guided run.

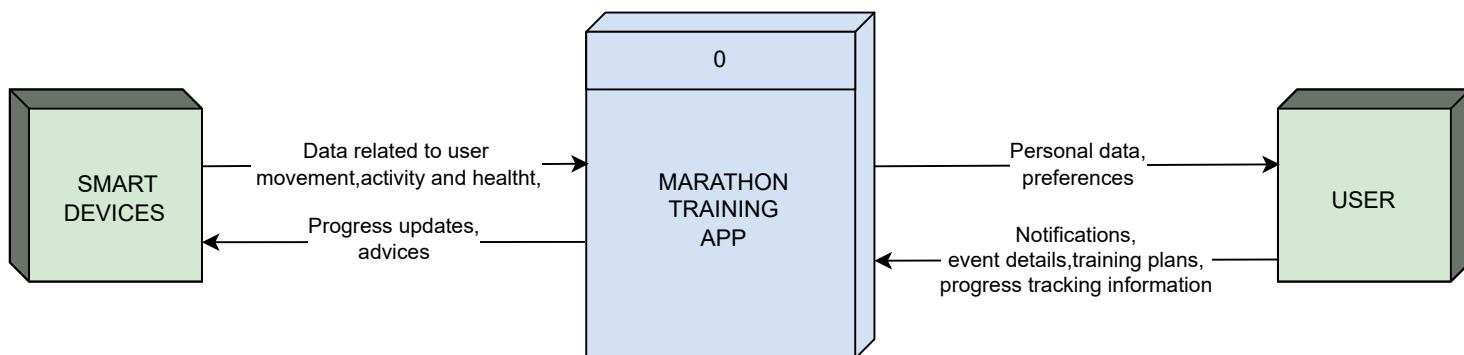
Another thing is that when we activate run through phone, the watch cannot track any distance and we need to hold our phone when running.

Please fix these issue and I guarantee your app will be the best!

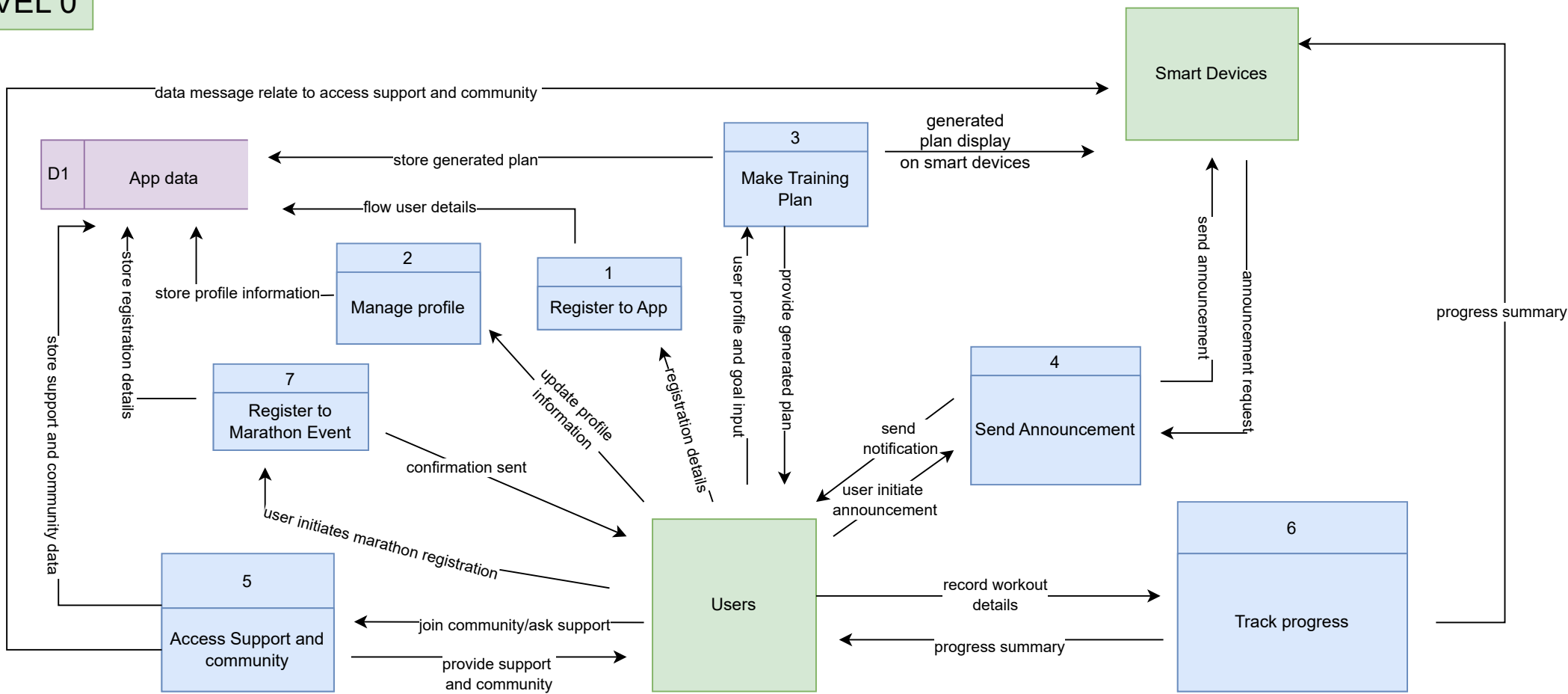
As can be seen from the above marathon apps reviews, it is clear that the map of the apps and its synchronization are not at the best. Thus, we gather all this information to avoid in our Marathon Training Application.

## 7.0 References

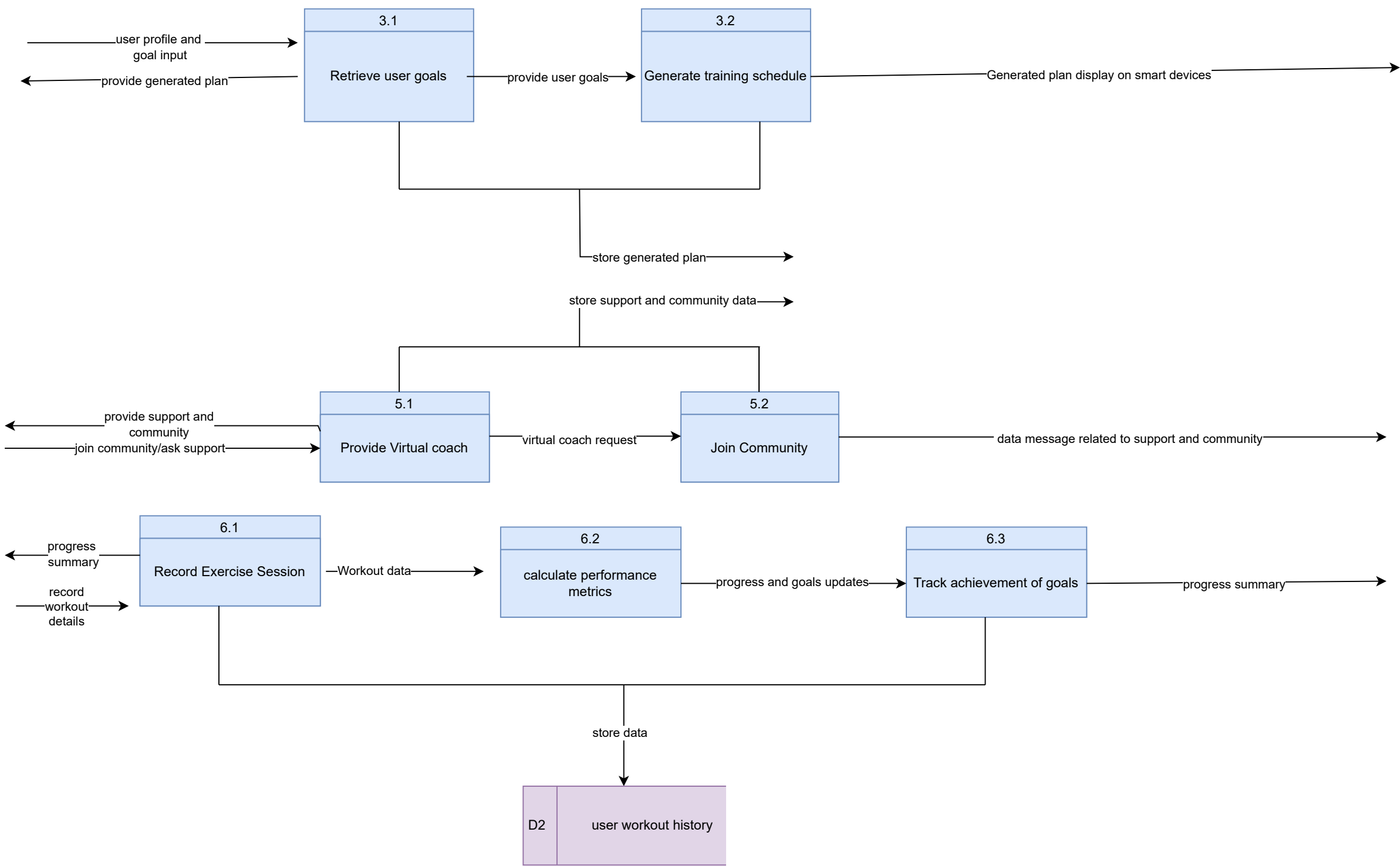
- Avram, M.-R., & Pop, F. (2022). Real-time running workouts monitoring using Cloud–Edge computing. *Neural Computing and Applications*. <https://doi.org/10.1007/s00521-021-06675-3>
- Feely, C., Caulfield, B., Lawlor, A., & Smyth, B. (2020). Using Case-Based Reasoning to Predict Marathon Performance and Recommend Tailored Training Plans. *Case-Based Reasoning Research and Development*, 67–81. [https://doi.org/10.1007/978-3-030-58342-2\\_5](https://doi.org/10.1007/978-3-030-58342-2_5)
- Janssen, M., Scheerder, J., Thibaut, E., Brombacher, A., & Vos, S. (2017). Who uses running apps and sports watches? Determinants and consumer profiles of event runners' usage of running-related smartphone applications and sports watches. *PLOS ONE*, 12(7), e0181167. <https://doi.org/10.1371/journal.pone.0181167>
- J. Lluch, M. Rebollo, Á. Calduch-Losa and R. Mollá, "Precision of Wearable GPS in Marathon Races," in IEEE Consumer Electronics Magazine, vol. 10, no. 1, pp. 32-38, 1 Jan. 2021, doi: 10.1109/MCE.2020.2986820.
- Pobiruchin, M., Suleder, J., Zowalla, R., & Wiesner, M. (2017). Accuracy and Adoption of Wearable Technology Used by Active Citizens: A Marathon Event Field Study. *JMIR MHealth and UHealth*, 5(2), e24. <https://doi.org/10.2196/mhealth.6395>
- Sun, C.-I., Huang, J.-T., Weng, S.-C., & Chien, M.-F. (2019). City Marathon Active Timing System Using Bluetooth Low Energy Technology. *Electronics*, 8(2), 252. <https://doi.org/10.3390/electronics8020252>



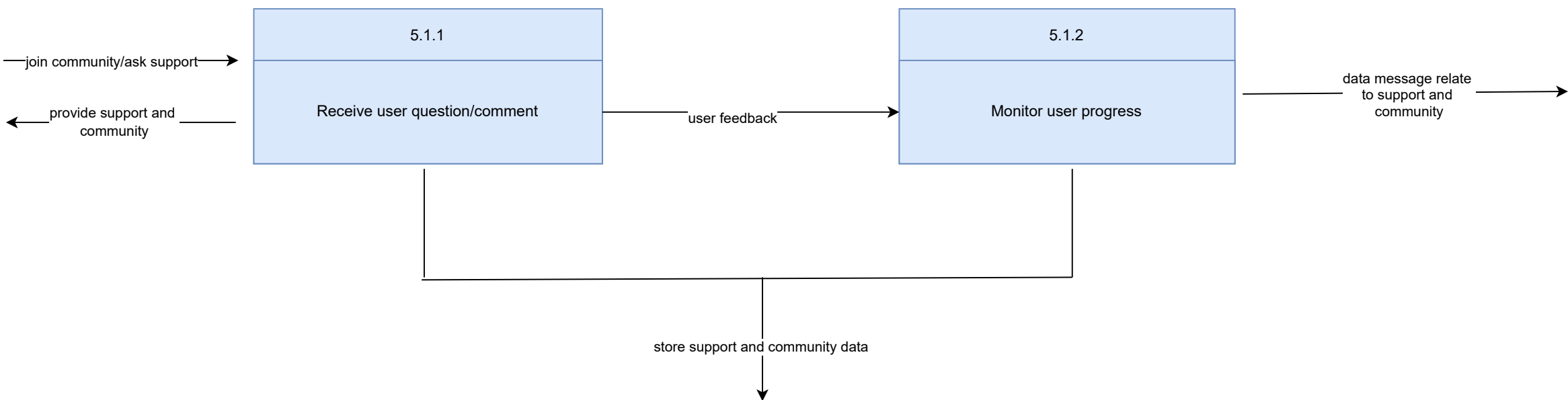
LEVEL 0



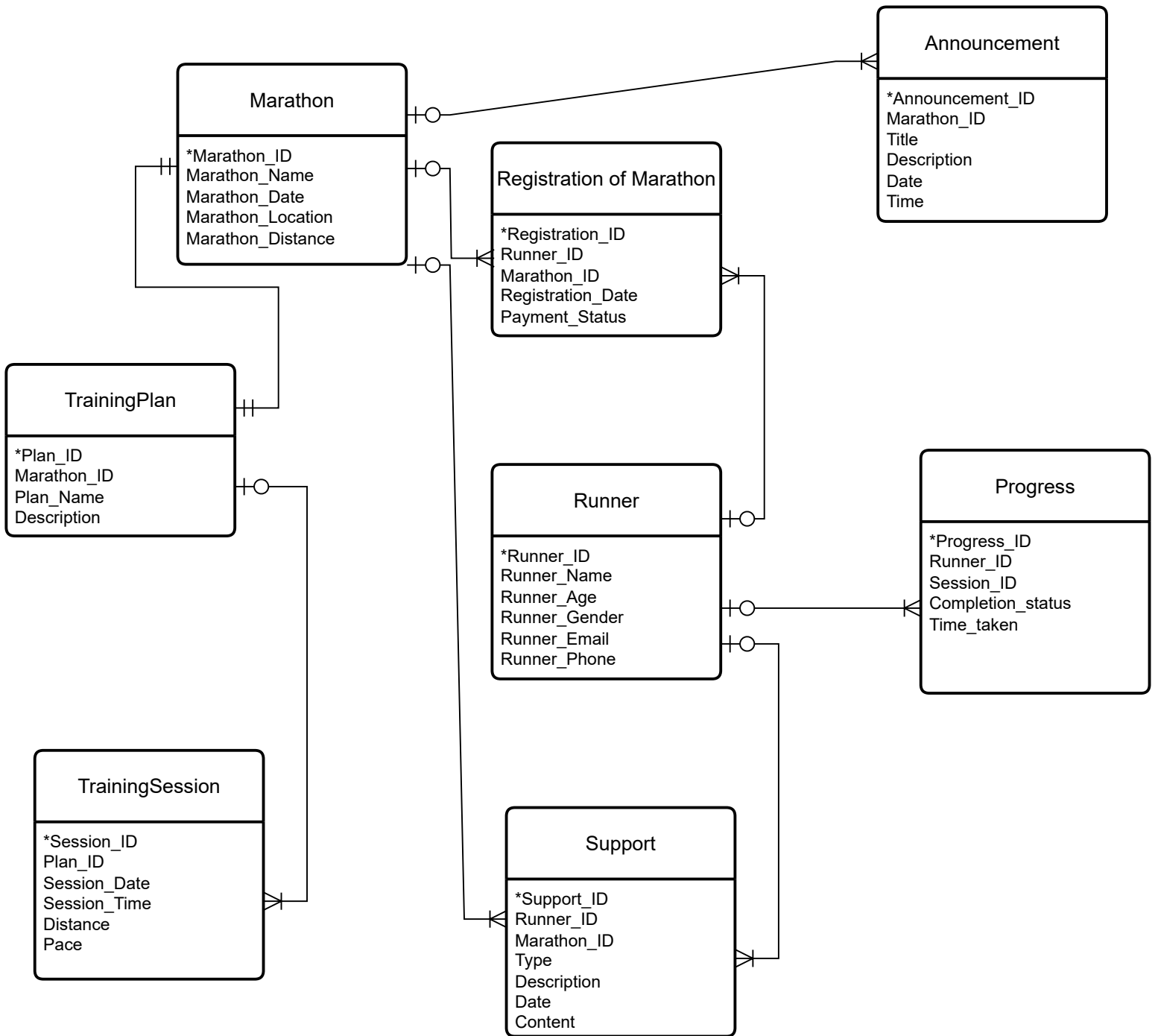
LEVEL 1

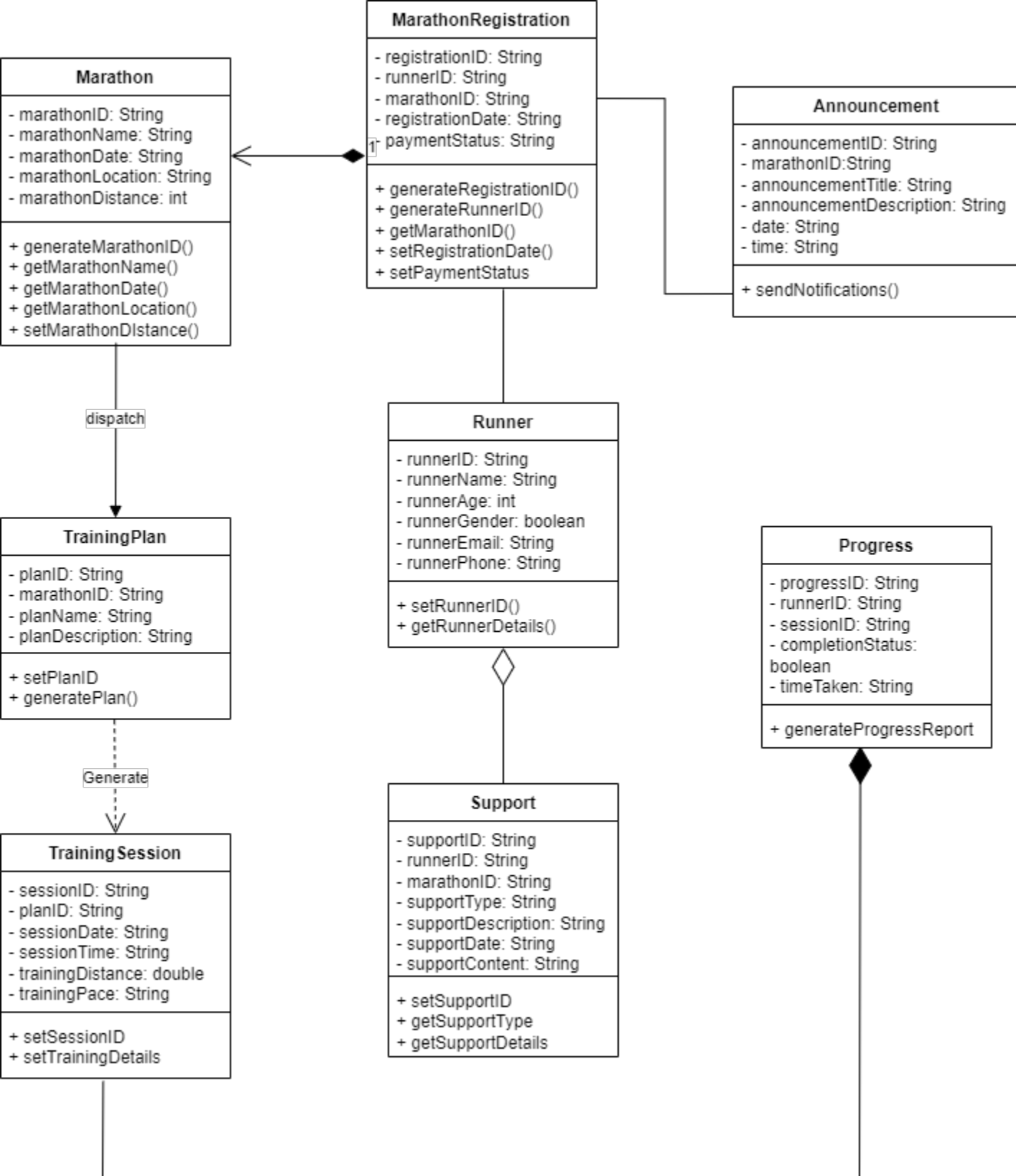


LEVEL 2









LOGIN



E-mail Address

Password

[Forgot Password?](#)

Sign In

Or

Connect with

Phone

E-mail

Other

Create an Account

SIGN UP



Username

E-mail Address

Password

☐ I agree to the [Terms & Conditions](#)

Sign Up

Or

Sign In Here

FORGOT PASSWORD

Forgot Password

E-mail Address

Verification Code

1

2

3

4

[Send Verification Code again](#)

New Password

Submit

Or

Sign In Here

USER GOALS

What is your primary marathon goal?

I want to run more this time

I want to join marathon events

I want to be fit and healthy

Next

USER GOALS

How familiar are you with marathon?

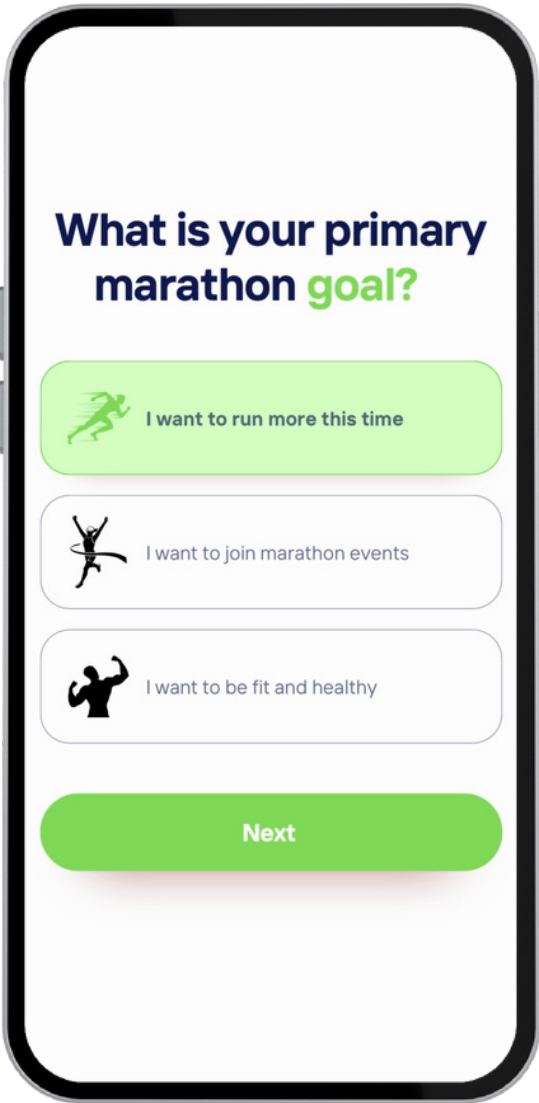
Not familiar

Somewhat familiar

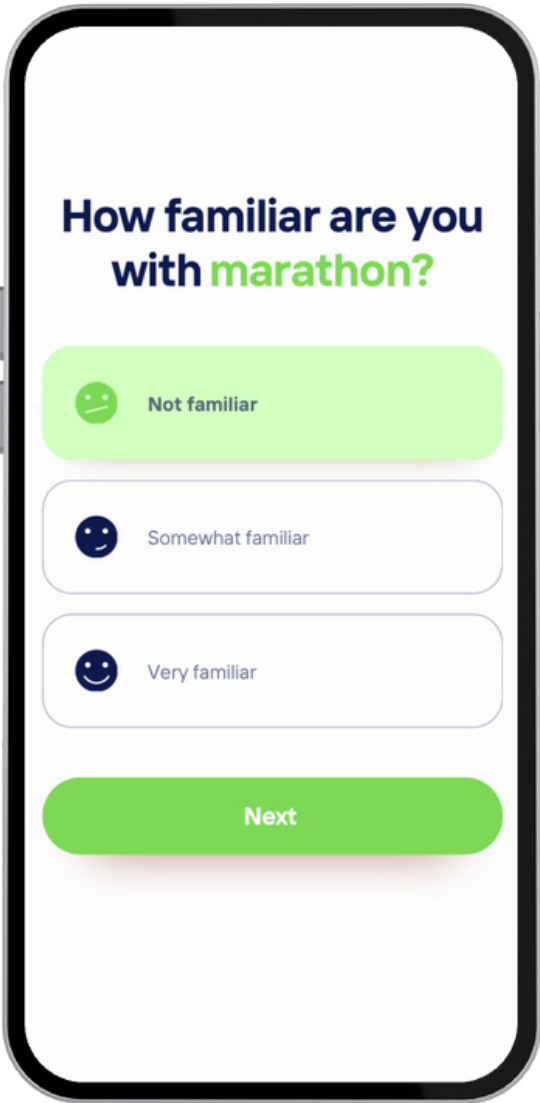
Very familiar

Next

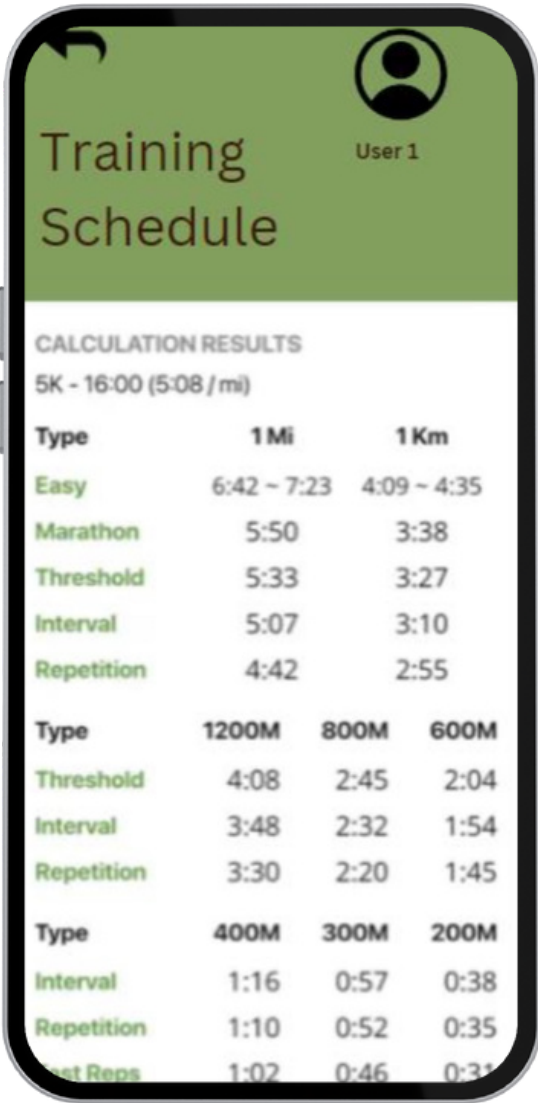
USER GOALS



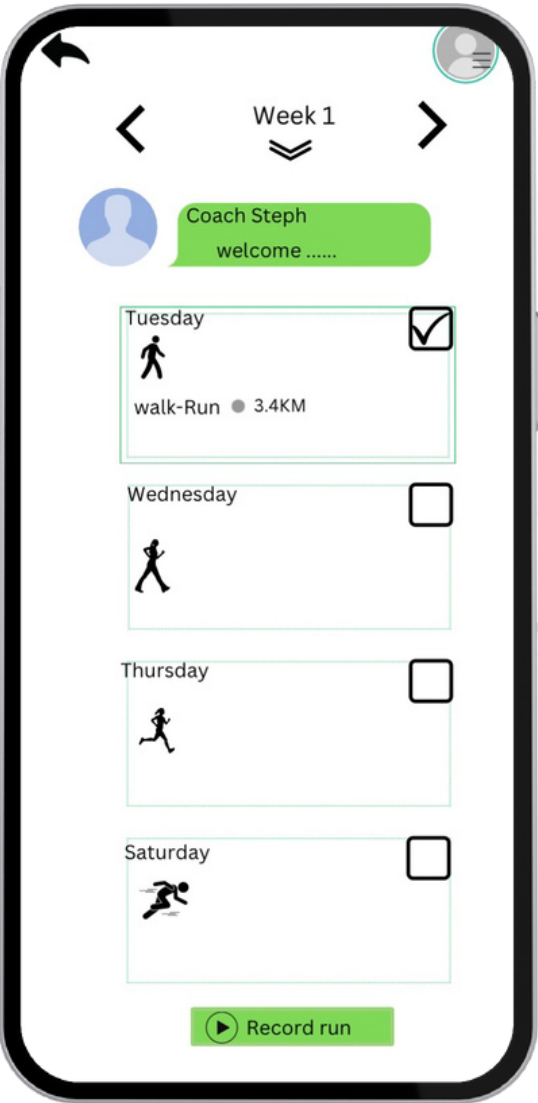
USER GOALS



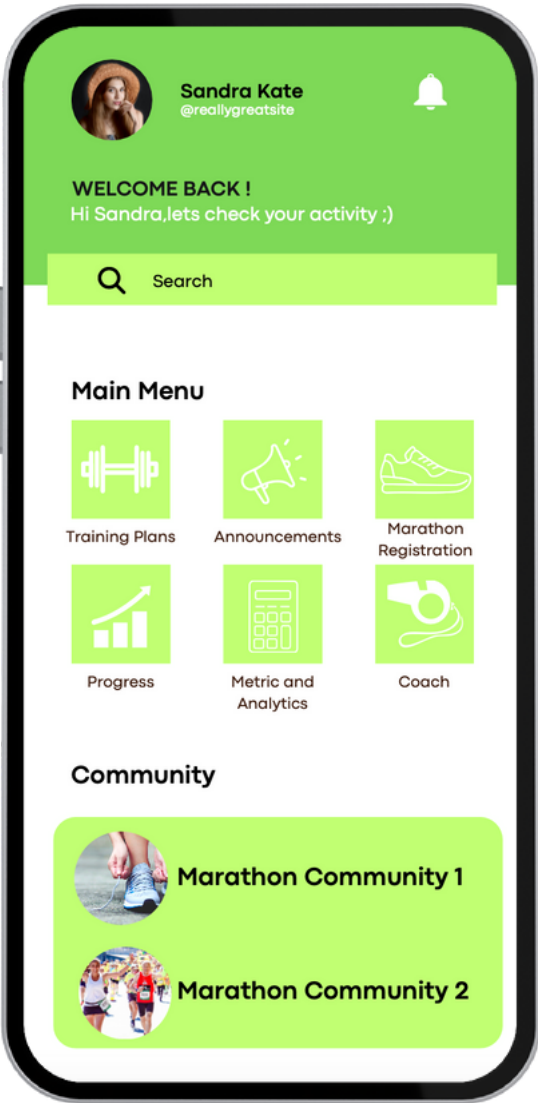
TRAINING SCHEDULE  
GENERATED



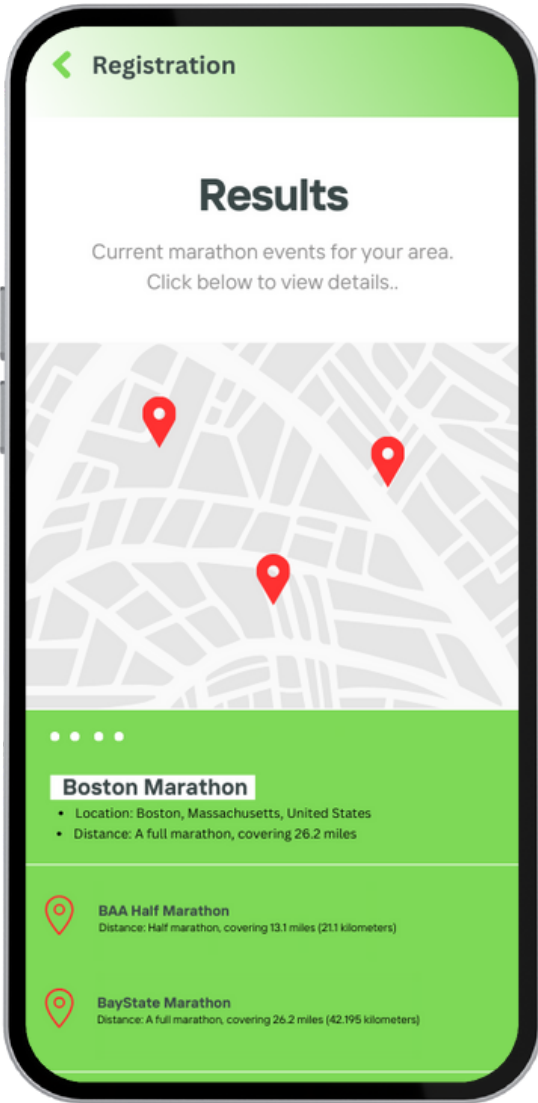
TRAINING CHECKED



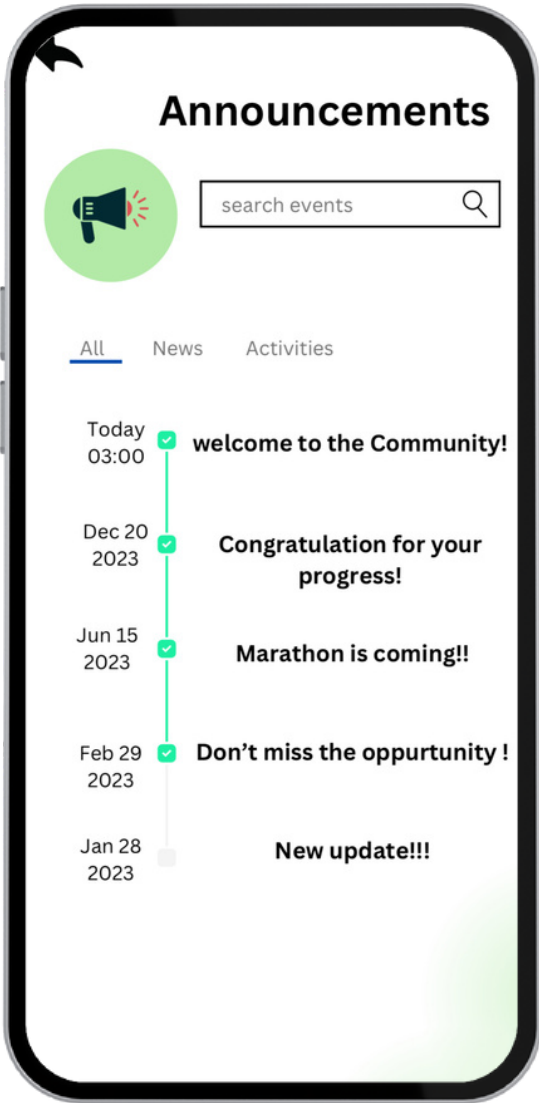
MAIN MENU



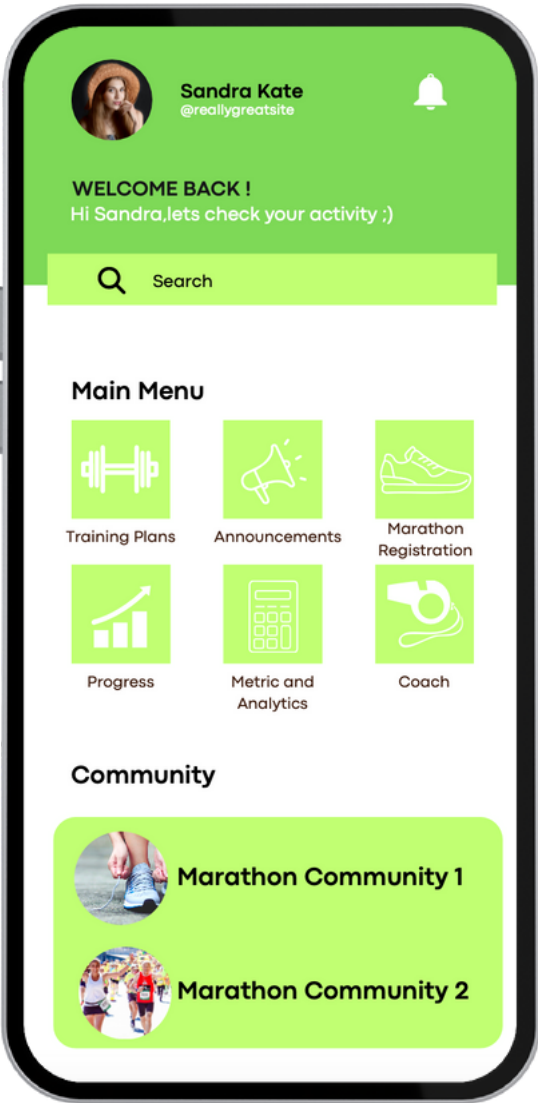
MARATHON REGISTRATION



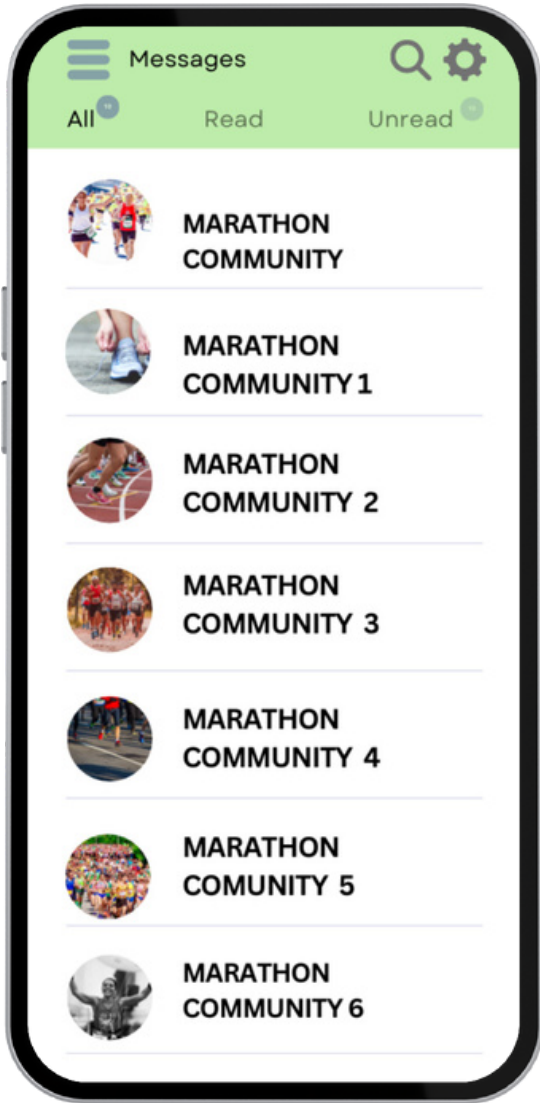
MARATHON ANNOUNCEMENTS



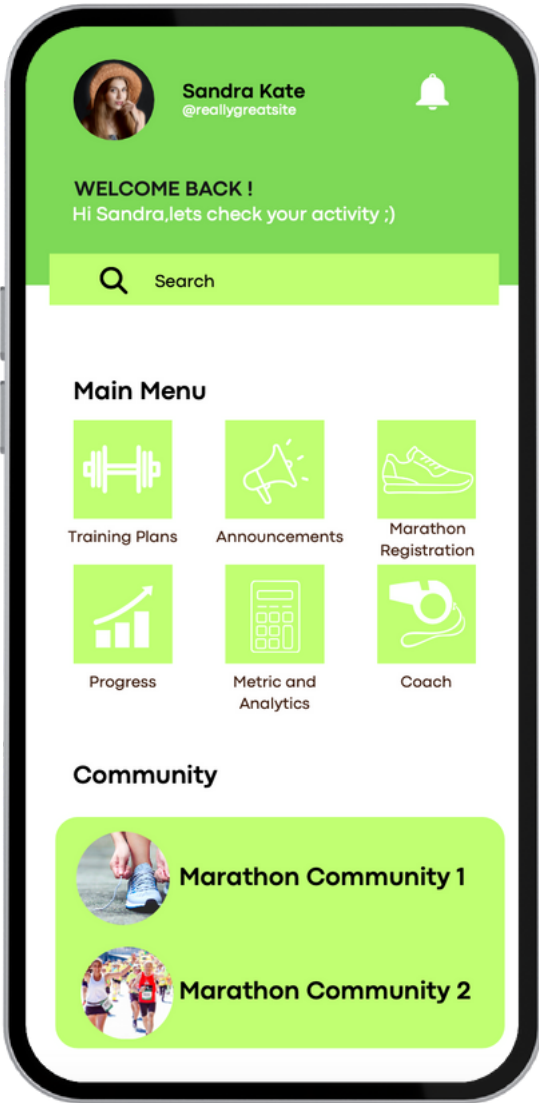
MAIN MENU



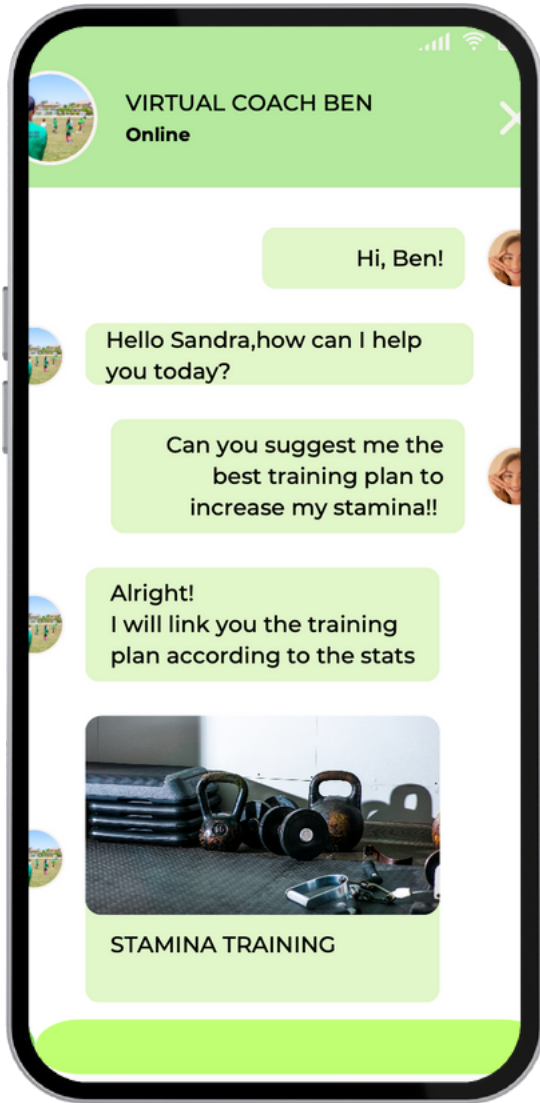
COMMUNITY SUPPORT



MAIN MENU

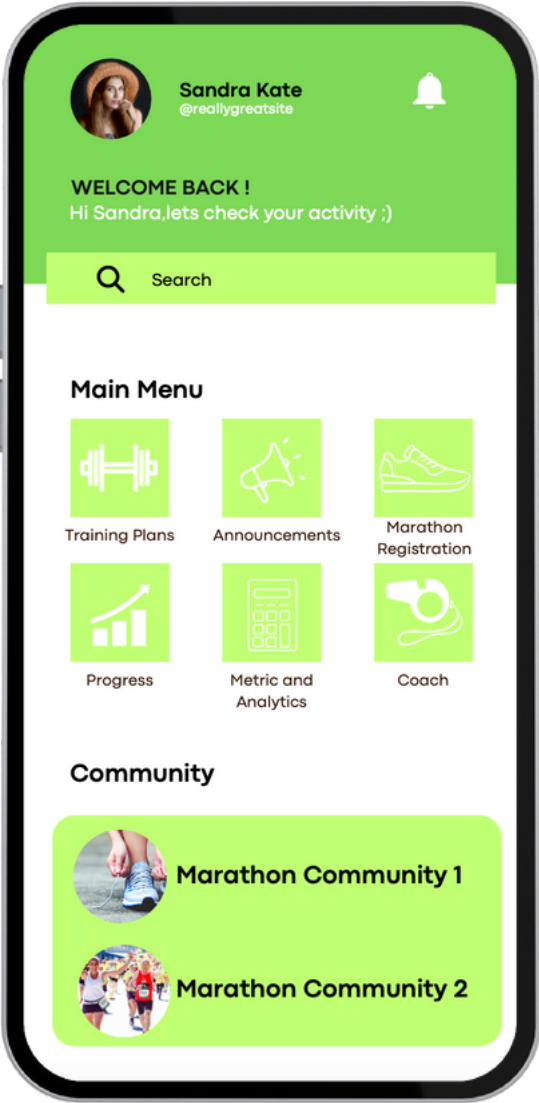


VIRTUAL COACH

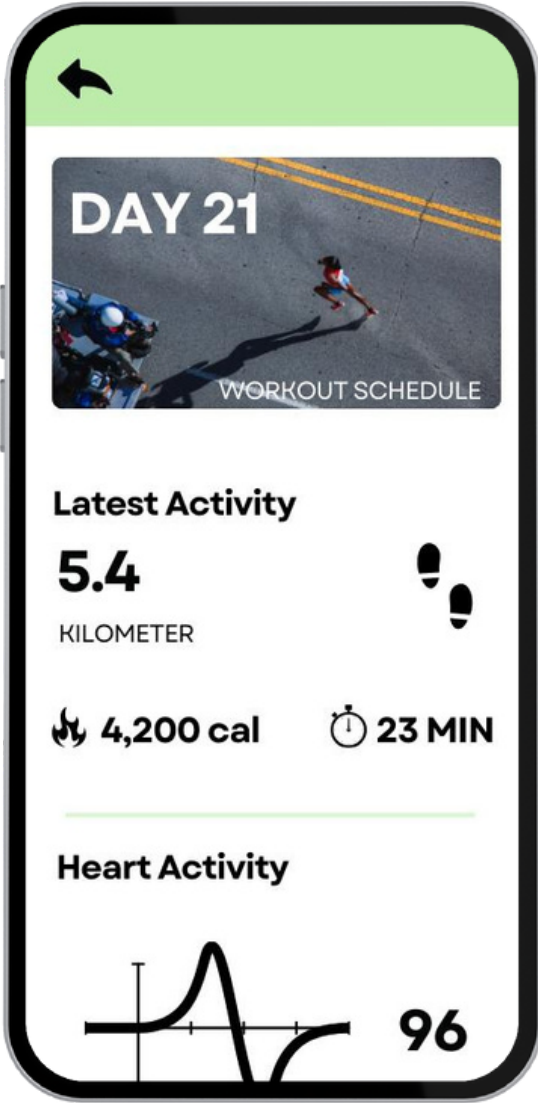




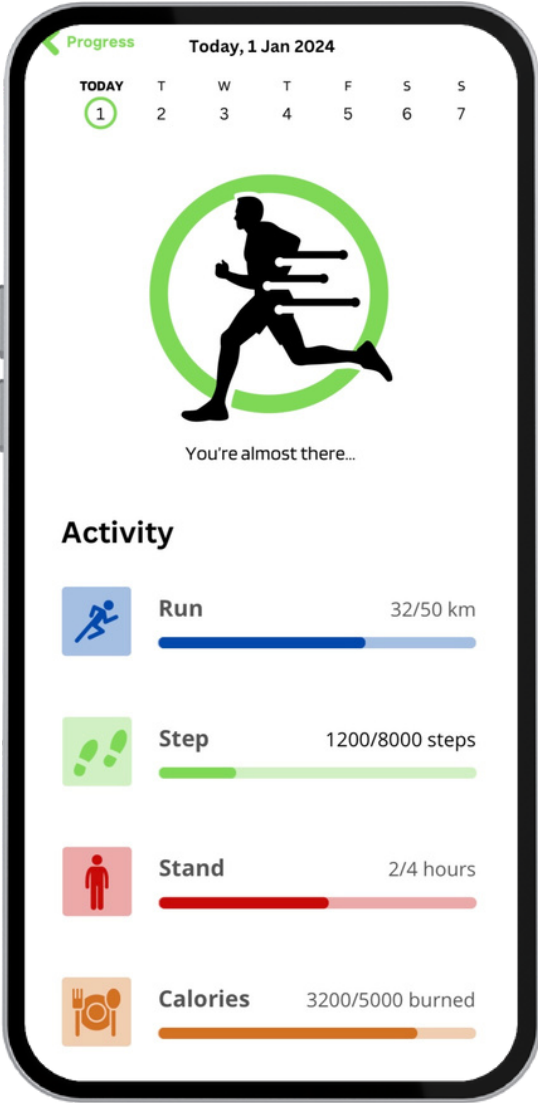
MAIN MENU



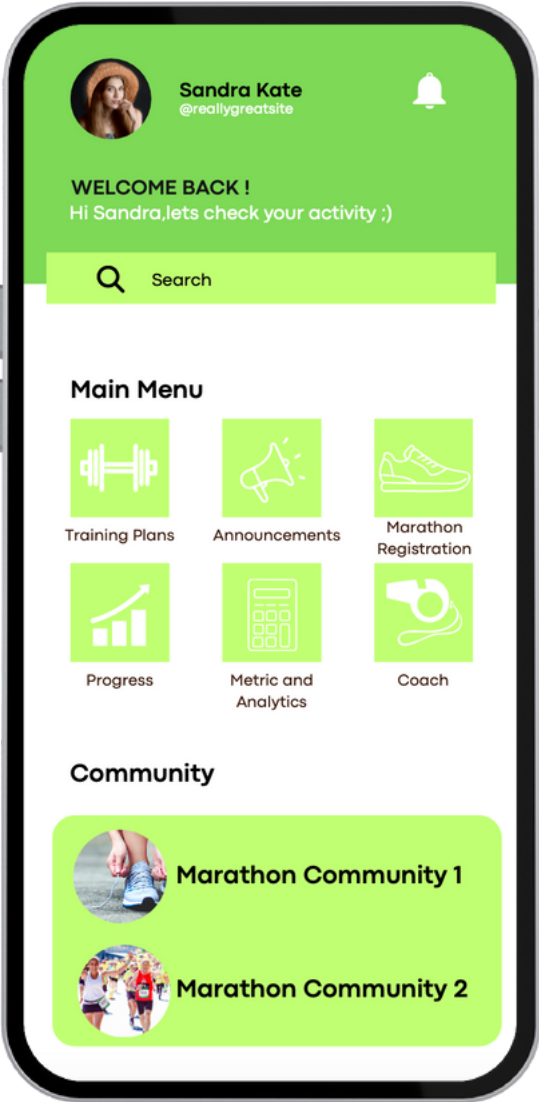
PROGRESS TRACKER



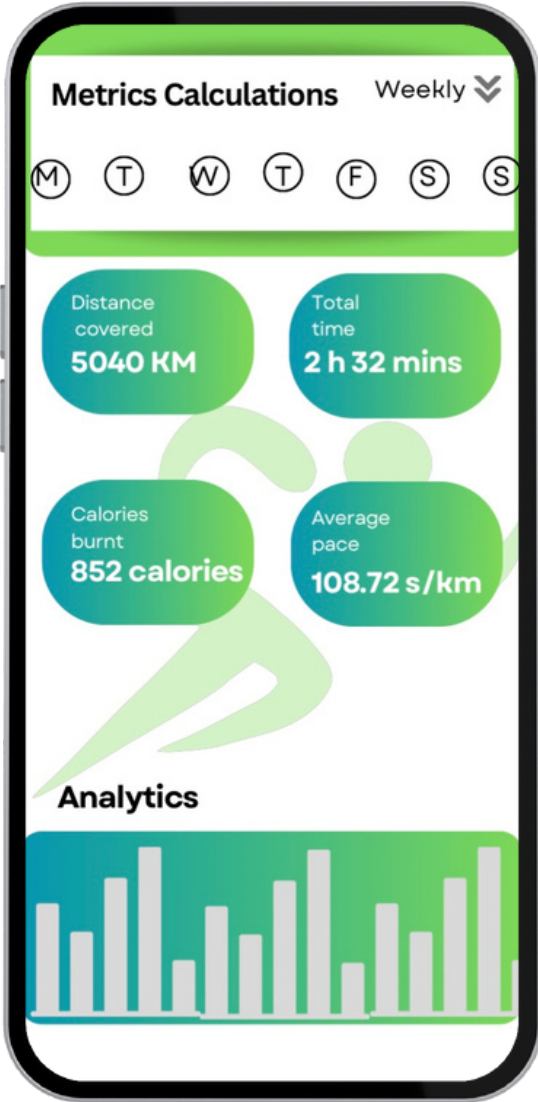
DETAILED PROGRESS



MAIN MENU



METRICS & PROGRESS SUMMARY



MARATHON ACHIEVEMENT

